

Instructions for ACL-IJCNLP 2021 Proceedings

Anonymous ACL-IJCNLP submission

Abstract

This is the report of the project for the Web Recommender System course held at the University of Copenhagen.

1 Week 6 [Feb 3 - Feb 9]: Familiarize Yourself with the Datasets

1.1 Dataset Description and Import

The dataset used in this project is the “Musical Instruments” dataset from the Amazon Review data 2023 (5-core subset). The dataset consists of 14,000 item ratings from nearly 900 users on approximately 500 items. The data is split into training and test sets, with 80% of the data allocated for training (“train.tsv”) and 20% for testing (“test.tsv”).

The dataset was imported using the Python library `pandas` and the `read_csv` function (Aho and Ullman, 1972).

1.2 Data Cleaning and Preprocessing

To ensure the quality of the dataset, we performed the following preprocessing steps:

- Removal of missing values (entries with missing user ID, item ID, or rating).
- Deduplication by keeping the most recent rating when a user rated the same item multiple times.
- Ensuring that all users in the test set are also present in the training set.

After cleaning, the dataset contained:

- Train set size: **XXX** ratings
- Test set size: **XXX** ratings

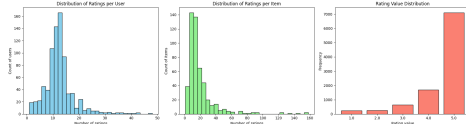


Figure 1: Distribution of Ratings per User, per Item, and Overall Rating Values

Item ID	High Ratings Count	Average Rating
ITEM1	XXX	XXX
ITEM2	XXX	XXX
ITEM3	XXX	XXX
ITEM4	XXX	XXX
ITEM5	XXX	XXX

Table 1: Top 5 Most Highly Rated Items

1.3 Exploratory Data Analysis (EDA) and Statistics

To gain insights into the dataset, we computed the following statistics:

- Distribution of ratings per user
- Distribution of ratings per item
- Overall rating value distribution

The histograms below visualize these distributions:

1.4 Highly Rated Items

To identify popular items, we computed the number of ratings ≥ 3 for each item and reported the top 5 most highly rated items:

1.5 Discussion

The dataset exhibits common characteristics of user-item interactions:

- There is a long-tail distribution where a small number of items receive most of the ratings, while many items have very few ratings.

- The majority of users have rated only a few items, which can lead to sparsity issues in collaborative filtering approaches.
- Certain items dominate the high-rated category, which may bias recommendation systems.

References

Alfred V. Aho and Jeffrey D. Ullman. 1972. *The Theory of Parsing, Translation and Compiling*, volume 1. Prentice-Hall, Englewood Cliffs, NJ.